

VG
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Requirements:

- Agitator driven by electric motor
 - ↳ smooth load
- Input power: 11 kW
- Input speed, $\omega_1 = 900 \text{ rpm}$
- Output speed, $\omega_2 = 245 - 255 \text{ rpm}$
 - ↳ Average = 250 rpm

Get the design power per strand (DPS)

$$DPS = \frac{11 \times 1}{1} = 11 \text{ kW}$$

Select chain size:

No. 40 roller chain with $N_1 = 40$

$$\text{Pitch} = 12.70 \text{ mm}$$

$$D_1 = \frac{P}{\sin\left(\frac{180}{N_1}\right)} = \frac{12.7}{\sin\left(\frac{180}{40}\right)} = 161.4 \text{ mm}$$

$D_1 < 400 \text{ mm}$, so it's fine.

Get N_2

$$\omega_2 = \frac{245 + 255}{2} = 250 \text{ rpm}$$

$$SR = \frac{\omega_1}{\omega_2} = \frac{900}{250} = 3.6$$

$$N_2 = N_1 (3.6) = 40 \times 3.6 = 144$$

Check $D_2 < 400 \text{ mm}$,

$$D_2 = \frac{P}{\sin \frac{180^\circ}{N_2}} = \frac{12.7}{\sin \left(\frac{180}{144} \right)} = 582.2 \text{ mm}$$

Since $D_2 > 400 \text{ mm}$, the chain is not suitable.

Since we are trying to minimise the chain size, we will use a double strand design instead.

Get the DPS again for 2 strand:

$$\text{DPS} = \frac{11 \times 1.0}{1.7} = 6.47 \text{ kW}$$

Select chain No. 40 roller chain with
 $N_1 = 25$

$$\text{Pitch} = 12.7 \text{ mm}$$

$$D_1 = \frac{P}{\sin \left(\frac{180}{N_1} \right)} = \frac{12.7}{\sin \left(\frac{180}{25} \right)} = 101.3298679 \text{ mm}$$

Since $D_1 < 400 \text{ mm}$, it is suitable.

$$\text{SR} = 3.6$$

$$N_2 = N_1 (\text{SR}) = 25(3.6) = 90$$

Since there is no sprocket in the catalogue with $N_2 = 90$, the chain is not suitable.

Re-select chain:

No. 50 chain with $N_1 = 13$

$$\text{Pitch} = 15.875 \text{ mm}$$

Since the minimum N_1 recommended is 17, we will use $N_1 = 17$.

$$\begin{aligned} N_2 &= N_1 (SR) \\ &= 17(3.6) \\ &= 61.2 \end{aligned}$$

$$D_1 = 86.39 \text{ mm}$$

From the catalogue, $N_2 = 60$, $D_2 = 304.33$

$$\begin{aligned} \omega_2 &= \frac{17}{60} \times 900 \\ &= 255 \text{ rpm} \end{aligned}$$

$$\therefore N_1 = 17, N_2 = 60$$

Get the tentative centre distance

$$\begin{aligned} \text{TCD} &= \frac{760}{P} \\ &= \frac{760}{15.875} \end{aligned}$$

$$\approx 47.9 \text{ pitches}$$

$$\begin{aligned}
 TCL &= 2TCD + \frac{N_2 + N_1}{2} + \frac{(N_2 - N_1)^2}{4\pi^2(TCD)} \\
 &= 2(47.9) + \frac{17+60}{2} + \frac{(60-17)^2}{4\pi^2(47.9)} \\
 &= 135.28 \text{ pitches}
 \end{aligned}$$

Get the closest even number of pitches that don't exceed 760mm.
 $L = 134$ pitches. Why use even pitches?

Get the centre distance: No offset link required.

$$\begin{aligned}
 C &= \frac{1}{4} \left\{ L - \frac{N_2 + N_1}{2} + \sqrt{\left[L - \frac{N_2 + N_1}{2} \right]^2 - \frac{8(N_2 - N_1)^2}{4\pi^2}} \right\} \\
 &= \frac{1}{4} \left\{ 134 - \frac{60+17}{2} + \sqrt{\left[134 - \frac{60+17}{2} \right]^2 - \frac{8(60-17)^2}{4\pi^2}} \right\} \\
 &= 47.25 \text{ pitches}
 \end{aligned}$$

$$\begin{aligned}
 C &= 47.25 \times 15.875 = 750.1640828 \text{ mm} \\
 &\approx 750.2 \text{ mm}
 \end{aligned}$$

Select lubrication:

Lubrication type B from the catalogue.

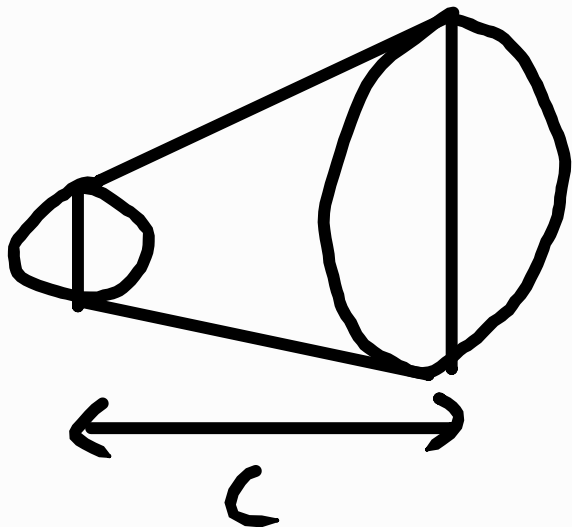
$$D_1 = 86.39 \text{ mm}$$

$$D_2 = 303.33 \text{ mm}$$

$$N_1 = 17$$

$$N_2 = 60$$

$$C = 750.2 \text{ mm}$$



$$\text{Area of the drive} = \frac{1}{2} \pi \left(\frac{D_1}{2} \right)^2 + \frac{1}{2} \pi \left(\frac{D_2}{2} \right)^2$$

$$+ \frac{1}{2} (D_1 + D_2) (C)$$

$$A_{\text{new}} = \frac{1}{2} \pi \left(\frac{86.39}{2} \right)^2 + \frac{1}{2} \pi \left(\frac{303.33}{2} \right)^2$$

$$+ \frac{1}{2} (86.39 + 303.33) (750.2)$$

$$A_{\text{new}} = 185.2466611 \times 10^3 \text{ mm}^2$$

Previous drive:

$$D_1 = 101.5 \text{ mm}$$

$$D_2 = 363.9 \text{ mm}$$

$$C = 750.4 \text{ mm}$$

$$\text{Area of the previous drive} = \frac{1}{2} \pi \left(\frac{D_1}{2} \right)^2 + \frac{1}{2} \pi \left(\frac{D_2}{2} \right)^2$$

$$+ \frac{1}{2} (D_1 + D_2) (C)$$

$$A_{\text{prev}} = \frac{1}{2} \pi \left(\frac{101.5}{2} \right)^2 + \frac{1}{2} \pi \left(\frac{363.9}{2} \right)^2$$

$$+ \frac{1}{2} (101.5 + 363.9) (750.4)$$

$$A_{\text{prev}} = 230.6662371 \times 10^3 \text{ mm}^2$$

Since $A_{\text{new}} < A_{\text{prev}}$, the new drive is indeed smaller than the previous drive.

Tabulate the two designs with all the particulars of the chains and sprockets for comparison.

Specify the sprocket types for both the driving and driven gears.

why use even pitches?

Odd pitches require the use of an offset link, which wear faster and should be avoided.